

Optimizing microdissection sample sizes for reliable microscale global metabolomics of host-microbiota metabolites in chicken gut samples

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Introduction

Metabolites play a key role in how microbiota influences host health. Here our goal was to develop a microscale global metabolomics method as part of the 3D'omics project, which aims to generate spatially resolved multi-omics information on a microscopic scale. Organs are flash frozen, subsegmented, carboxymethylcellulose (CMC) embedded, and cryosectioned and further microdissected using a laser-microdissection microscope. Analyzing the metabolome of gut microsections enables investigation of host-microbiota interactions at a microscopic level, especially when combined with (meta)genomic and (meta)transcriptomic data.

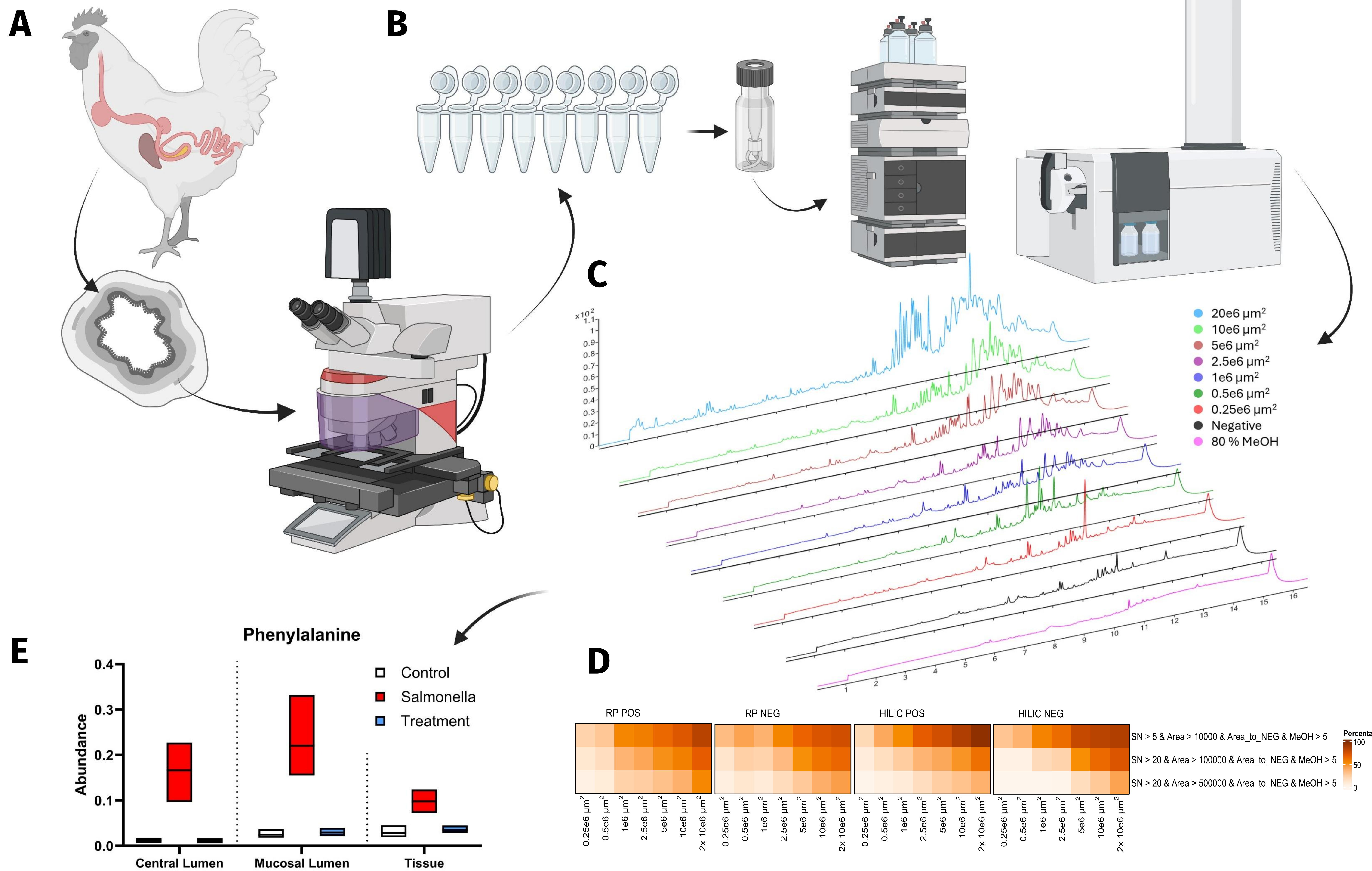


Figure: Methods and Results. (A) We evaluated 10 µm thick CMC embedded laser microdissected chicken caecum samples, sample size ranging from 0.25e6 to 10e6 µm². We analyzed samples from a trial evaluating mitigation potential of a symbiotic, multispecies feed additive (PoltryStar®) on the effects of *Salmonella* Infantis infection (Drauch et al. 2025). (B) Metabolite extraction was carried out with 15 µl of 80 % (v:v) methanol and the samples were analyzed with both RP and HILIC chromatography coupled with Agilent 6546 QTOF. A data quality assessment was conducted based on peak area, signal-to-noise ratio, and sample-to-negative control and sample-to-solvent blank ratios. (C) Examples of total ion chromatograms from RP POS mode. (D) Percentages of signals from identified metabolites fulfilling the data quality criteria (SN, signal to noise; NEG, negative sample control [microdissected CMC sample cut next to tissue]; MeOH, methanol blank). (E) Example showing how phenylalanine levels (signal abundance normalized to sample area) differ between different parts of the caecum and how treatment can mitigate some of the effects of the *Salmonella* Infantis infection.

Conclusions

As the microdissection samples become smaller, the quality of the signals detected decreases limiting usability of the data. However, these results do show that it is possible, for example, to measure metabolite profiles separately from the host tissue and the central and mucosal lumen from chicken gut microsamples to detect biologically meaningful differences. These results are vital for optimizing sizes of microsection samples in future studies, balancing the need for detailed locational data with the limitations of metabolite detection to ensure robust and reliable microscale global metabolomics results that can be combined with other omics data measured at microscopic level.



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References:
Drauch et al. Poltry Science, 2025
<https://doi.org/10.1016/j.psj.2025.104890>